

Eodalmantina

EE-o-dal-man-ee-tee-na

- **When** 465 million years ago (Middle Ordovician)
- **Fossil location** France, Portugal, Spain
- **Habitat** Seafloor
- **Length** Up to 1½ in (4 cm)

Like many trilobites, *Eodalmantina* had large eyes and good vision. Trilobites were among the first animals to evolve sophisticated eyes. These consisted of many tiny, crystalline lenses packed together in a honeycomb pattern, like the compound eyes of insects. *Eodalmantina* had distinctive bean-shaped eyes. Its long body tapered toward the tail, which was tipped by a short spine.



Ceratarges

SER-a-tar-gees

- **When** 380–359 million years ago (Middle to Late Devonian)
- **Fossil location** Morocco
- **Habitat** Seafloor
- **Length** 2½ in (6.6 cm)

Ceratarges was one of many trilobites with spectacular spines and horns. These prickly weapons may have been used to ward off predators. Another theory, however, is that they evolved as a result of battles between rival trilobites fighting over mates, like the antlers of modern stag beetles.



Encrinurus

EN-crine-yoo-rus

- **When** 444 million years ago (Silurian)
- **Fossil location** Worldwide
- **Habitat** Seafloor
- **Length** Up to 2 in (5 cm)

This small trilobite had many berry-shaped bumps on the shield protecting its head. *Encrinurus*'s eyes were probably situated at the end of short stalks. It may have spent a lot of time hiding in the mud on the seabed, with only its eyes above the surface.



Phacops

FAY-cops

- **When** 380–359 million years ago (Middle to Late Devonian)
- **Fossil location** Worldwide
- **Habitat** Seafloor
- **Length** Up to 2¼ in (6 cm)

Named for its keen sense of sight, *Phacops* ("lens eye"), like *Eodalmantina*, had bulging eyes that gave it good vision, suggesting it lived in well-lit areas such as shallow seas. One of the most common and widespread trilobites, *Phacops* has been found in Europe, Africa, Australia, and North America. Geologists even use its fossils as a handy way to estimate the age of a rock.



Lenses in eye

► **ROLLING UP**
Phacops could curl up in a tight ball to protect its softer undersides when attacked, much like some modern woodlice.

